Activity 14: Data Wrangling, Visualization, and Reflection

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Armed Forces Data Wrangling Redux

Load packages

Chunk 1

```r
# Load required packages
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.1     v stringr   1.5.2
v ggplot2   4.0.0     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.1.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becon
```

```r
library(janitor)
```

```
Warning: package 'janitor' was built under R version 4.5.2
```

```
Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```r
library(readxl)
```

Load the data

Chunk 2

```r
# Load the Armed Forces data set
armed_raw <- read_excel("US_Armed_Forces_(6_2025).xlsx", skip = 1)
```

```
New names:
* `` -> `...1`
* `` -> `...3`
* `` -> `...4`
* `` -> `...6`
* `` -> `...7`
* `` -> `...9`
* `` -> `...10`
* `` -> `...12`
* `` -> `...13`
* `` -> `...15`
* `` -> `...16`
* `` -> `...18`
* `` -> `...19`
```

View armed_raw

Chunk 3

```r
# View head of armed_raw data
head(armed_raw)
```

```
# A tibble: 6 x 19
  ...1      Army    ...3    ...4    Navy  ...6  ...7  `Marine Corps` ...9  ...10
  <chr>     <chr>   <chr>   <chr>   <chr> <chr> <chr> <chr>          <chr> <chr>
1 Pay Grade Male    Female  Total   Male  Fema~ Total Male           Fema~ Total
2 E1        7429.0  1326.0  8755.0  8903~ 3434~ 1233~ 7849.0         655.0 8504~
3 E2        22338.0 4336.0  26674.0 1750~ 5833~ 2333~ 15034.0        1684~ 1671~
4 E3        43775.0 10229.0 54004.0 2543~ 9103~ 3453~ 35239.0        4174~ 3941~
5 E4        79234.0 15143.0 94377.0 3385~ 9959~ 4381~ 28519.0        2961~ 3148~
6 E5        54803.0 10954.0 65757.0 5814~ 1616~ 7431~ 22262.0        2670~ 2493~
# i 9 more variables: `Air Force` <chr>, ...12 <chr>, ...13 <chr>,
#   `Space Force` <chr>, ...15 <chr>, ...16 <chr>, Total <chr>, ...18 <chr>,
#   ...19 <chr>
```

Extract header row and view it

Chunk 4

```r
header_row <- read_excel("US_Armed_Forces_(6_2025).xlsx", n_max = 1)
```

```
New names:
* `` -> `...2`
* `` -> `...3`
* `` -> `...4`
* `` -> `...5`
* `` -> `...6`
* `` -> `...7`
* `` -> `...8`
* `` -> `...9`
* `` -> `...10`
* `` -> `...11`
* `` -> `...12`
* `` -> `...13`
* `` -> `...14`
* `` -> `...15`
* `` -> `...16`
* `` -> `...17`
```

```r
header_row
```

```
# A tibble: 1 x 17
  Active-Duty Personnel ~1 ...2  ...3  ...4  ...5  ...6  ...7  ...8  ...9  ...10
```

```
  <lgl>                      <chr> <lgl> <lgl> <chr> <lgl> <lgl> <chr> <lgl> <lgl>
1 NA                         Army  NA    NA    Navy  NA    NA    Mari~ NA    NA
# i abbreviated name:
#   1: `Active-Duty Personnel by Service Branch, Sex, and Pay Grade`
# i 7 more variables: ...11 <chr>, ...12 <lgl>, ...13 <lgl>, ...14 <chr>,
#   ...15 <lgl>, ...16 <lgl>, ...17 <chr>
```

Clean names

Chunk 5

```r
cleaned <- armed_raw %>%
  clean_names()
names(cleaned)
```

```
 [1] "x1"           "army"         "x3"           "x4"           "navy"
 [6] "x6"           "x7"           "marine_corps" "x9"           "x10"
[11] "air_force"    "x12"          "x13"          "space_force"  "x15"
[16] "x16"          "total"        "x18"          "x19"
```

Rename column names

Chunk 6

```r
cleaned <- armed_raw %>%
  clean_names() %>%
  rename(
    pay_grade = x1,
    army_male = army,
    army_female = x3,
    army_total = x4,
    navy_male = navy,
    navy_female = x6,
    navy_total = x7,
    marine_male = marine_corps,
    marine_female = x9,
    marine_total = x10,
    airforce_male = air_force,
    airforce_female = x12,
    airforce_total = x13,
    space_male = space_force,
```

```
    space_female = x15,
    space_total = x16,
    total_overall = total
)
```

Data wrangling

Chunk 7

```
group_df <- cleaned |>
  select(pay_grade, matches("_(male|female|total)$")) |>
  pivot_longer(
    cols = -pay_grade,
    names_to = c("branch", "gender"),
    names_sep = "_",
    values_to = "count"
  ) |>
  filter(gender != "total") |>
  mutate(count = as.numeric(count)) |>
  filter(!is.na(count))
```

```
Warning: There was 1 warning in `mutate()`.
i In argument: `count = as.numeric(count)`.
Caused by warning:
! NAs introduced by coercion
```

Final data set

Chunk 8

```
head(group_df)
```

```
# A tibble: 6 x 4
  pay_grade branch gender count
  <chr>     <chr>  <chr>  <dbl>
1 E1        army   male    7429
2 E1        army   female  1326
3 E1        navy   male    8903
4 E1        navy   female  3434
5 E1        marine male    7849
6 E1        marine female   655
```

Baby Names Visualization

For the names I chose Michael and Emily, they are among the most consistently popular names in the US. I can get a trend using these 2 names which would allow me to explore the long-term popularity and compare how the trends have changed for male and female names. The plot I choose is Line plot, this is because it will give us a comparison of the trend between the 2 names from 1880 to the present.

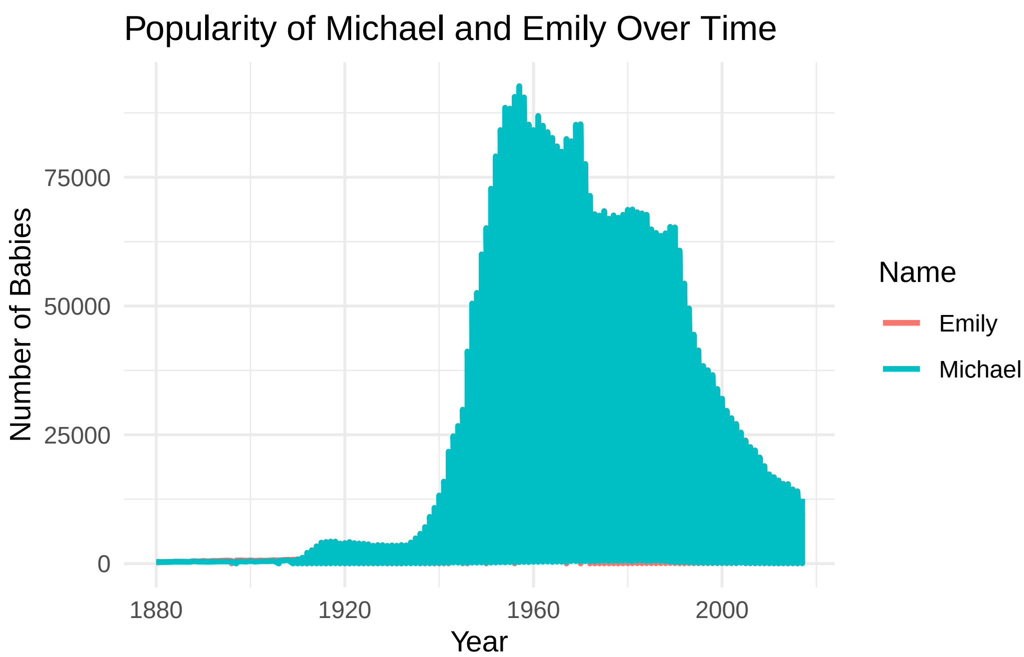
Chunk 9

```r
library(babynames)
```

```
Warning: package 'babynames' was built under R version 4.5.2
```

```r
library(dplyr)
library(ggplot2)
selected_names <- babynames %>%
  filter(name %in% c("Michael", "Emily"))
ggplot(selected_names, aes(x = year, y = n, color = name)) +
  geom_line(size = 1) +
  labs(
    title = "Popularity of Michael and Emily Over Time",
    x = "Year",
    y = "Number of Babies",
    color = "Name"
    ) +
    theme_minimal()
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```



Plotting a Mathematical Function

For calculating the volume of a box with respect to the side length that is cut out, we will have to use the below function

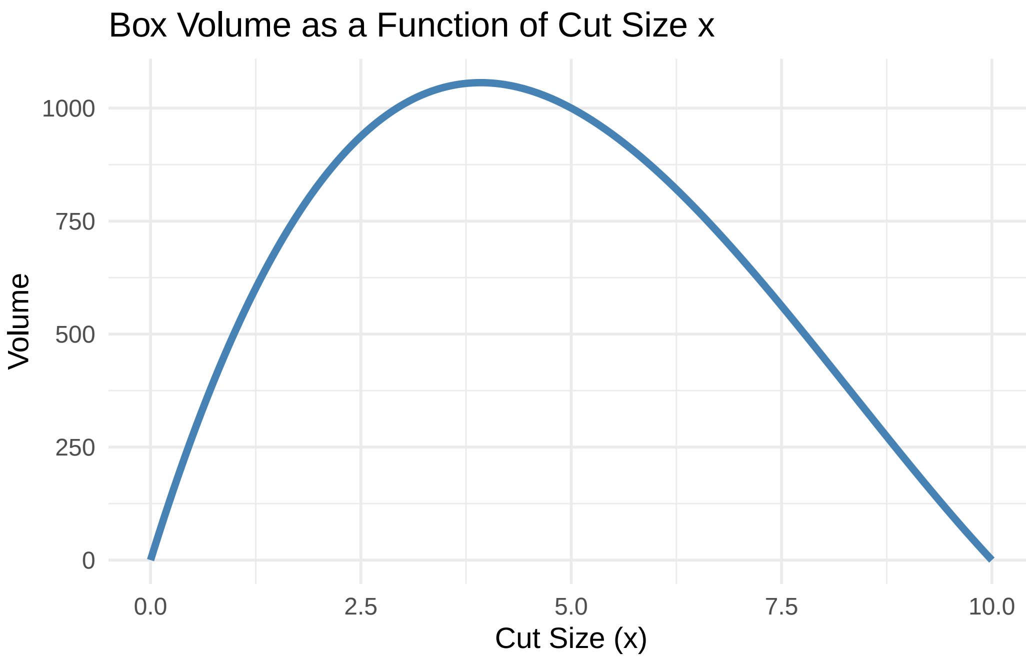
V(x) = x * (30-2x) * (20-2x)

The code below is for the plot that shows how the Volume is affected when the value of x increases

Chunk 10

```
# Define the volume function
volume <- function(x) {
  x * (30 - 2*x) * (20 - 2*x)
}
# Create a sequence of x values from 0 to 10
x_vals <- seq(0, 10, length.out = 300)
vol_vals <- volume(x_vals)
# Plot
library(ggplot2)
ggplot(data.frame(x = x_vals, volume = vol_vals), aes(x, volume)) +
  geom_line(color = "steelblue", linewidth = 1.3) +
```

```
labs(
  title = "Box Volume as a Function of Cut Size x",
  x = "Cut Size (x)",
  y = "Volume"
) +
theme_minimal()
```



What I feel I have learned

I have learnt that there are many things to take into consideration when we want to make our code or work open to all and reproducible, such as it should be able to run on anyone's workstation. Based on learning in R, I have learnt that it is best to keep rechecking your code so that it follows all the steps of data wrangling so that we get the best output. I have learnt how to use clean_name() and uncount() in a proper sense. I have come from a state of understnading the code I am using to writing it part by part by myself.

Code Appendix

```r
# Load required packages
library(tidyverse)
library(janitor)
library(readxl)
```

```r
# Load the Armed Forces dataset
armed_raw <- read_excel("US_Armed_Forces_(6_2025).xlsx", skip = 1)
```

```
New names:
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* `` -> `...3`
* `` -> `...4`
* `` -> `...6`
* `` -> `...7`
* `` -> `...9`
* `` -> `...10`
* `` -> `...12`
* `` -> `...13`
* `` -> `...15`
* `` -> `...16`
* `` -> `...18`
* `` -> `...19`
```

```r
# View head of armed_raw data
head(armed_raw)
```

```
# A tibble: 6 x 19
  ...1      Army    ...3    ...4    Navy  ...6  ...7  `Marine Corps` ...9  ...10
  <chr>     <chr>   <chr>   <chr>   <chr> <chr> <chr> <chr>          <chr> <chr>
1 Pay Grade Male    Female  Total   Male  Fema~ Total Male           Fema~ Total
2 E1        7429.0  1326.0  8755.0  8903~ 3434~ 1233~ 7849.0         655.0 8504~
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# i 9 more variables: `Air Force` <chr>, ...12 <chr>, ...13 <chr>,
#   `Space Force` <chr>, ...15 <chr>, ...16 <chr>, Total <chr>, ...18 <chr>,
#   ...19 <chr>
```

```r
header_row <- read_excel("US_Armed_Forces_(6_2025).xlsx", n_max = 1)
```

```
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* `` -> `...8`
* `` -> `...9`
* `` -> `...10`
* `` -> `...11`
* `` -> `...12`
* `` -> `...13`
* `` -> `...14`
* `` -> `...15`
* `` -> `...16`
* `` -> `...17`
```

```r
header_row
```

```
# A tibble: 1 x 17
  Active-Duty Personnel ~1 ...2  ...3  ...4  ...5  ...6  ...7  ...8  ...9  ...10
  <lgl>                    <chr> <lgl> <lgl> <chr> <lgl> <lgl> <chr> <lgl> <lgl>
1 NA                       Army  NA    NA    Navy  NA    NA    Mari~ NA    NA
# i abbreviated name:
#   1: `Active-Duty Personnel by Service Branch, Sex, and Pay Grade`
# i 7 more variables: ...11 <chr>, ...12 <lgl>, ...13 <lgl>, ...14 <chr>,
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cleaned <- armed_raw %>%
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```
 [1] "x1"           "army"         "x3"           "x4"           "navy"
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[11] "air_force"    "x12"          "x13"          "space_force"  "x15"
[16] "x16"          "total"        "x18"          "x19"
```

```r
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    army_female = x3,
    army_total = x4,
    navy_male = navy,
    navy_female = x6,
    navy_total = x7,
    marine_male = marine_corps,
    marine_female = x9,
    marine_total = x10,
    airforce_male = air_force,
    airforce_female = x12,
    airforce_total = x13,
    space_male = space_force,
    space_female = x15,
    space_total = x16,
    total_overall = total
)
```

```r
group_df <- cleaned |>
  select(pay_grade, matches("_(male|female|total)$")) |>
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    cols = -pay_grade,
    names_to = c("branch", "gender"),
    names_sep = "_",
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  filter(gender != "total") |>
  mutate(count = as.numeric(count)) |>
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```r
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```
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2 E1        army   female  1326
3 E1        navy   male    8903
4 E1        navy   female  3434
5 E1        marine male    7849
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```r
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library(ggplot2)
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ggplot(selected_names, aes(x = year, y = n, color = name)) +
  geom_line(size = 1) +
  labs(
    title = "Popularity of Michael and Emily Over Time",
    x = "Year",
    y = "Number of Babies",
    color = "Name"
    ) +
    theme_minimal()
```
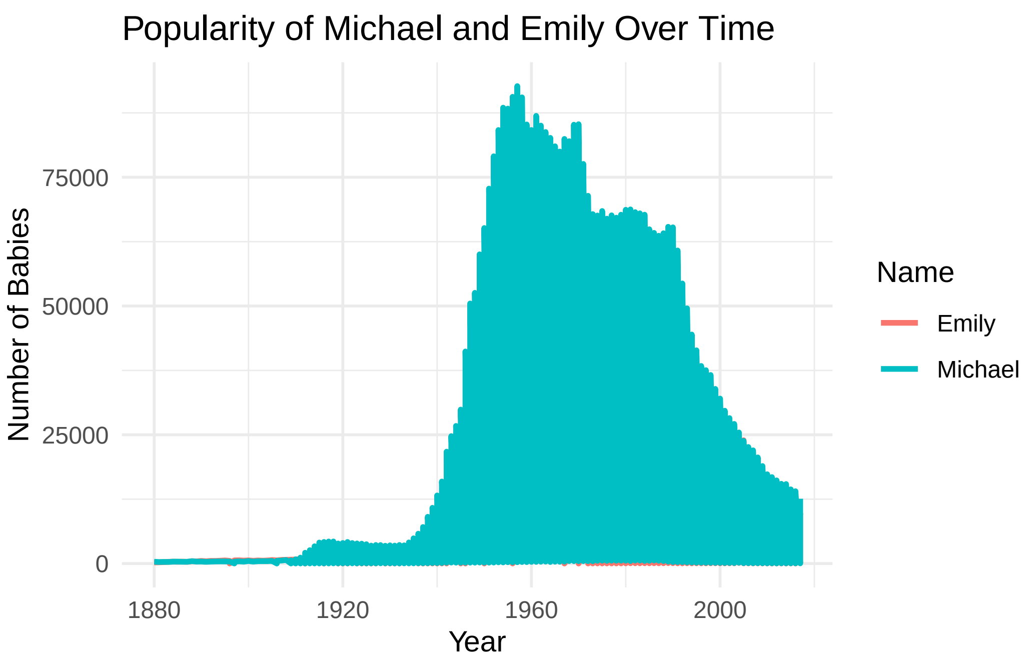
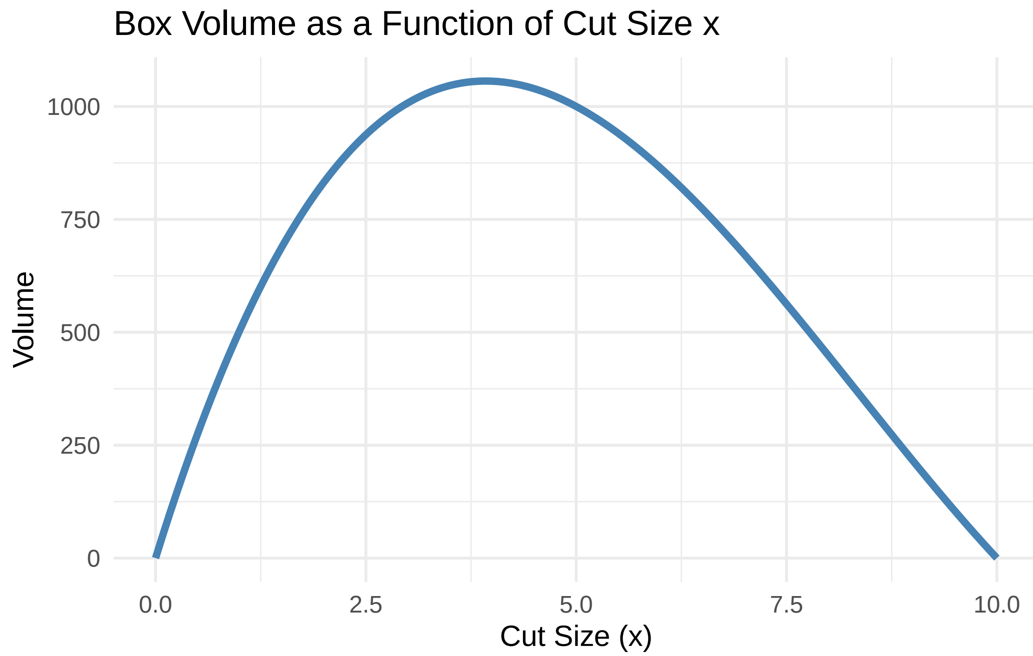




```r
# Define the volume function
volume <- function(x) {
  x * (30 - 2*x) * (20 - 2*x)
}
# Create a sequence of x values from 0 to 10
x_vals <- seq(0, 10, length.out = 300)
vol_vals <- volume(x_vals)
# Plot
library(ggplot2)
ggplot(data.frame(x = x_vals, volume = vol_vals), aes(x, volume)) +
  geom_line(color = "steelblue", linewidth = 1.3) +
  labs(
    title = "Box Volume as a Function of Cut Size x",
    x = "Cut Size (x)",
    y = "Volume"
  ) +
  theme_minimal()
```


Box Volume as a Function of Cut Size x
Volume
1000
750
500
250
0
0.0
2.5
5.0
7.5
10.0
Cut Size (x)